STORAGE

AWS provides cloud storage services for storing, accessing, and managing data according to different application requirements. These services allow organizations to store data without maintaining physical storage infrastructure and support workloads ranging from application files to long-term archives.

AWS storage services differ in how data is organized, accessed, and shared. Selecting an appropriate service depends on factors such as data type, access frequency, performance requirements, availability, and storage cost.

Amazon Elastic Block Store (Amazon EBS)

Amazon Elastic Block Store (Amazon EBS) provides **block-level storage** for use with Amazon Elastic Compute Cloud (Amazon EC2) instances. EBS stores data in volumes that can be attached to EC2 instances and used similarly to storage drives in a physical computer.

EBS volumes provide persistent storage, meaning that data can remain available independently of the running state of an EC2 instance. They are commonly used for operating systems, applications, databases, and other workloads that require consistent storage performance.

Amazon EBS Volumes

An **EBS volume** is a storage resource that can be created and attached to a compatible EC2 instance. After attachment, the volume can be formatted with a file system and used to store application files or other data.

EBS volumes are created within a specific Availability Zone. They can be attached to compatible EC2 instances in the same Availability Zone, subject to the volume type and attachment limitations.

An EBS volume can serve as the root volume of an EC2 instance, storing the operating system and files required to run the instance. Additional EBS volumes can be attached to provide separate storage for application data, such as database files.

An EBS volume can also be detached from one instance and attached to another compatible instance within the same Availability Zone. This allows data to be retained and reused without recreating the storage resource.

Amazon EBS Volume Types

Amazon EBS provides different volume types to support varying storage performance requirements. These are organized into two main categories:

- **Solid State Drive (SSD) volumes** – Designed for workloads that require frequent read and write operations with low latency. They are commonly used for boot volumes, transactional databases, and applications that require responsive storage.
- **Hard Disk Drive (HDD) volumes** – Designed for throughput-intensive workloads that process large amounts of sequential data. They are suitable for applications such as log processing and large-scale data processing.

The appropriate volume type depends on the workload's performance requirements, including input/output operations per second (IOPS), throughput, and latency.

PROPERTY OF STI

Amazon EBS Snapshots

An **EBS snapshot** is a point-in-time backup of an EBS volume. Snapshots can be used to protect stored data and create new EBS volumes when recovery or duplication is required.

EBS snapshots are incremental by default. After the initial snapshot, subsequent snapshots store the blocks that have changed since the previous snapshot, helping reduce the amount of additional backup storage required.

EBS snapshots are stored in Amazon S3 infrastructure managed by AWS and can be used to create new volumes in different Availability Zones within the same Region. To create a volume in another Region, the snapshot must first be copied to that Region.

Benefits of Amazon EBS

Amazon EBS provides several benefits for applications running on Amazon EC2:

- Persistent storage – Data can remain available when an instance is stopped or when a volume is detached, subject to the configured deletion settings.
- Flexible capacity – Storage capacity and supported performance settings can be modified as workload requirements change.
- Storage performance options – SSD and HDD volume types support different workload requirements.
- Data protection – Snapshots provide a way to back up and restore volume data.
- Encryption – EBS supports encryption of volumes and snapshots to help protect stored data.

Amazon Simple Storage Service (Amazon S3)

Amazon Simple Storage Service (Amazon S3) is an **object storage service** that allows users to store and retrieve data from the AWS Cloud. Unlike block storage, which organizes data into volumes, Amazon S3 stores data as objects within containers called buckets.

Amazon S3 is designed to store large amounts of data and supports various types of files, including documents, images, videos, application backups, and website assets. It is commonly used for data backup, content storage, static website hosting, and data analytics.

Amazon S3 Buckets and Objects

Amazon S3 organizes data using buckets and objects. A **bucket** is a container for storing objects, while an **object** consists of the stored data and its associated metadata.

The main components of Amazon S3 storage include:

- **Buckets** – Containers used to organize and store objects. Each general purpose bucket has a globally unique name within an AWS partition and is created in a specific AWS Region.
- **Objects** – Individual files or data stored in a bucket, together with their associated metadata.
- **Object keys** – Unique identifiers used to locate objects within a bucket. Each object is identified by its key within the bucket.
- **Metadata** – Information associated with an object, such as its content type and other properties.

PROPERTY OF STI
PROPERTY OF STI
PROPERTY OF STI

For example, an organization can create a bucket for its website assets and store images, documents, and other files as separate objects. Each object can be accessed using its corresponding key and appropriate permissions.

Amazon S3 Storage Classes

Amazon S3 provides different storage classes to accommodate varying data access patterns and storage requirements. These classes differ in cost, availability characteristics, retrieval options, and minimum storage duration requirements.

Common Amazon S3 storage classes include:

- S3 Standard – Designed for frequently accessed data that requires high availability and low-latency access.
- S3 Intelligent-Tiering – Automatically moves eligible objects between access tiers based on changing access patterns to help optimize storage costs.
- S3 Standard-Infrequent Access (S3 Standard-IA) – Designed for data that is accessed less frequently but requires rapid retrieval when needed.
- S3 One Zone-Infrequent Access (S3 One Zone-IA) – Stores data in a single Availability Zone and is suitable for infrequently accessed data that can be recreated or is protected by another copy.
- S3 Glacier Instant Retrieval – Designed for rarely accessed archival data that still requires retrieval in milliseconds.
- S3 Glacier Flexible Retrieval – Designed for archival data that can tolerate retrieval times ranging from minutes to hours.
- S3 Glacier Deep Archive – Designed for long-term archival data that is rarely accessed and can tolerate retrieval times measured in hours.

Selecting a storage class depends on how frequently the data is accessed, how quickly it must be retrieved, and the cost requirements of the workload.

Amazon S3 Versioning

S3 Versioning allows multiple versions of an object to be maintained within the same bucket. When versioning is enabled, uploading an object with the same key creates a new version rather than permanently replacing the existing version.

Versioning helps protect data against accidental overwriting and deletion. Previous versions can be retrieved when needed, making it useful for applications that require recovery from unintended changes.

Versioning must be enabled at the bucket level. Once enabled, it can be suspended, but the bucket cannot return to an unversioned state.

Amazon S3 Lifecycle Management

Amazon S3 Lifecycle allows users to define rules that automatically manage objects throughout their storage lifecycle. These rules can transition eligible objects to different storage classes or expire objects when they are no longer required.

PROPERTY OF STI
PROPERTY OF STI
PROPERTY OF STI

Lifecycle rules can perform actions such as:

- Transitioning objects – Moving eligible objects to lower-cost storage classes as their access requirements change.
- Expiring objects – Automatically deleting objects after a specified period.
- Managing previous versions – Transitioning or expiring noncurrent object versions when versioning is enabled.
- Removing incomplete uploads – Aborting incomplete multipart uploads after a specified number of days.

For example, an organization can configure a lifecycle rule to transition older backup files to an archival storage class and delete them after the required retention period.

Amazon S3 Security and Access Management

Amazon S3 provides security features that help control access to stored data and protect objects from unauthorized use.

Common security mechanisms include:

- **IAM policies** – Define permissions that allow or deny users and roles access to Amazon S3 resources.
- **Bucket policies** – Resource-based policies that control access to buckets and their objects.
- **S3 Block Public Access** – Settings that help prevent unintended public access to buckets and objects.
- **Encryption** – Protects stored data through server-side encryption. Amazon S3 automatically encrypts new object uploads by default.
- **Access logging** – Provides records of requests made to buckets when server access logging is configured.

Access permissions should follow the principle of least privilege, granting only the permissions required to perform authorized tasks.

Benefits of Amazon S3

Amazon S3 provides several benefits for storing and managing data in the AWS Cloud:

- Scalability – Supports storing large amounts of data without requiring users to provision storage capacity in advance.
- Durability – Designed for 99.999999999% (11 nines) durability of objects across its storage classes.
- Storage flexibility – Provides storage classes for different access patterns and retention requirements.
- Data protection – Supports versioning, encryption, and access controls to help protect stored objects.
- Automated management – Lifecycle rules help manage data retention and storage costs.

Amazon Elastic File System (Amazon EFS)

Amazon Elastic File System (Amazon EFS) is a managed file storage service that provides shared file storage for applications running in the AWS Cloud. Unlike block storage, which provides storage volumes for individual computing resources, Amazon EFS uses a file system that can be accessed by multiple computing resources simultaneously.

PROPERTY OF STI
PROPERTY OF STI
PROPERTY OF STI

Amazon EFS supports the Network File System (NFS) protocol and is designed for Linux-based workloads. It is commonly used for applications that require shared access to files, such as web applications, content management systems, and data processing workloads.

Amazon EFS File Systems

An **EFS file system** is a storage resource that organizes data into files and directories. Applications can access these files using standard file system operations, such as creating, reading, modifying, and deleting files.

Amazon EFS can be mounted on multiple compatible Amazon EC2 instances simultaneously. This allows applications running on different instances to access the same files without maintaining separate copies.

Amazon EFS automatically grows and shrinks as files are added or removed, reducing the need to provision and manage storage capacity manually.

Amazon EFS Storage Classes

Amazon EFS provides storage classes that support different data access patterns and cost requirements. These classes allow organizations to balance storage performance and cost based on how frequently files are accessed.

The main Amazon EFS storage classes include:

- EFS Standard – Designed for frequently accessed files that require low-latency access.
- EFS Infrequent Access (EFS IA) – Designed for files that are accessed less frequently and can tolerate higher access latency.
- EFS Archive – Designed for files that are rarely accessed and can tolerate higher access latency in exchange for lower storage costs.

Amazon EFS can use lifecycle management policies to move files between eligible storage classes based on their access patterns.

Amazon EFS Performance and Throughput

Amazon EFS provides performance and throughput options to support different application requirements.

Two important aspects of EFS performance are:

- **Performance** – Amazon EFS supports General Purpose performance mode, which is designed for applications that require low-latency file operations.
- **Throughput** – Amazon EFS provides Elastic Throughput, Provisioned Throughput, and Bursting Throughput options to accommodate different levels of file system activity.

Elastic Throughput automatically adjusts throughput based on workload activity. Provisioned Throughput allows users to specify a throughput level, while Bursting Throughput provides throughput based on the amount of data stored and available burst credits.

The appropriate throughput option depends on the application's access patterns and performance requirements.

PROPERTY OF STI
PROPERTY OF STI
PROPERTY OF STI

Amazon EFS Availability and Scalability
Amazon EFS supports file storage across AWS Availability Zones, depending on the file system configuration.

Amazon EFS provides two primary file system types:

- **Regional file systems** – Store data redundantly across multiple Availability Zones within an AWS Region to support high availability and durability.
- **One Zone file systems** – Store data within a single Availability Zone and are suitable for workloads that do not require multi-AZ storage resilience.

Amazon EFS can scale automatically as storage requirements change. Multiple compatible computing resources can access the same file system, allowing applications to share data as their computing capacity increases.

Amazon EFS Security and Access Management
Amazon EFS provides security features to help protect stored files and control access to file systems.

Common security mechanisms include:

- **IAM policies** – Control permissions for managing EFS resources and can authorize supported client access operations.
- **Security groups** – Control network traffic between file system mount targets and connected computing resources.
- **Encryption at rest** – Protects data stored in an EFS file system.
- **Encryption in transit** – Protects data transferred between supported clients and the file system.
- **EFS access points** – Provide application-specific entry points that can enforce a root directory and POSIX user identity for file access.

These mechanisms can be combined with file system permissions to manage access to shared files.

Benefits of Amazon EFS
Amazon EFS provides several benefits for applications that require shared file storage:

- Shared file access – Allows multiple compatible computing resources to access the same files simultaneously.
- Automatic scaling – Adjusts storage capacity as files are added or removed.
- Managed infrastructure – Reduces the need to provision and maintain file storage servers.
- Storage flexibility – Provides storage classes and throughput options for different workload requirements.
- Data protection – Supports encryption, access controls, and multi-AZ storage configurations for workloads requiring higher availability.

Amazon S3 Glacier

Amazon S3 Glacier is a set of low-cost archival storage classes within Amazon Simple Storage Service (Amazon S3). These storage classes are designed for data that is accessed infrequently but must be retained for extended periods.

PROPERTY OF STI
PROPERTY OF STI
PROPERTY OF STI

Amazon S3 Glacier storage classes are commonly used for long-term backups, historical records, compliance archives, and disaster recovery data. They allow organizations to reduce storage costs by selecting retrieval options that match how frequently and how quickly archived data must be accessed.

Amazon S3 Glacier Storage Classes

Amazon S3 provides three Glacier storage classes with different retrieval characteristics:

- S3 Glacier Instant Retrieval – Designed for rarely accessed data that requires immediate access. Objects can be retrieved in milliseconds.
- S3 Glacier Flexible Retrieval – Designed for archival data that does not require immediate access. It provides retrieval options ranging from minutes to hours.
- S3 Glacier Deep Archive – Designed for long-term archival data that is accessed very rarely. It offers low-cost storage with retrieval times measured in hours.

These storage classes support different retention and retrieval requirements. Organizations should consider the expected access frequency, retrieval time, and storage costs when selecting an archival storage class.

Amazon S3 Glacier Retrieval Options

Retrieving archived objects depends on the storage class and the urgency of the request.

For S3 Glacier Flexible Retrieval, the available retrieval options include:

- Expedited retrieval – Provides access to archived data in approximately 1–5 minutes, subject to capacity availability.
- Standard retrieval – Typically completes within 3–5 hours.
- Bulk retrieval – Typically completes within 5–12 hours and is suitable for retrieving large amounts of archived data at a lower cost.

For S3 Glacier Deep Archive, retrieval options include:

- Standard retrieval – Typically completes within 12 hours.
- Bulk retrieval – Typically completes within 48 hours.

Objects in S3 Glacier Instant Retrieval can be accessed directly without initiating a restore operation. For Flexible Retrieval and Deep Archive, a restore request is generally required before the archived object can be accessed.

Amazon S3 Glacier Data Management

Amazon S3 Lifecycle rules can automatically transition eligible objects from other S3 storage classes to Glacier storage classes as their access requirements change.

For example, an organization can store recently created backup files in S3 Standard and configure a lifecycle rule to transition them to S3 Glacier Flexible Retrieval when they are no longer accessed regularly.

Lifecycle rules can also expire archived objects after their required retention period. However, Glacier storage classes have minimum storage duration requirements. Deleting or transitioning objects before the required duration ends may result in early deletion charges.

PROPERTY OF STI
PROPERTY OF STI
PROPERTY OF STI

Amazon S3 Glacier Security and Durability

Amazon S3 Glacier storage classes use Amazon S3 security and data protection mechanisms to help protect archived information.

These include:

- Encryption – Protects archived objects through server-side encryption.
- Access control – Uses IAM policies and bucket policies to manage permissions for archived objects.
- Durability – Designed for 99.999999999% (11 nines) durability of stored objects.
- Versioning – Supports retaining previous versions of objects when S3 Versioning is enabled.
- S3 Object Lock – Can prevent object versions from being deleted or overwritten for a specified retention period when properly configured.

These features help organizations protect archived data while meeting applicable retention and access requirements.

Benefits of Amazon S3 Glacier

Amazon S3 Glacier storage classes provide several benefits for long-term data storage:

- Lower storage costs – Offer cost-effective options for data that is accessed infrequently.
- Flexible retrieval – Provide different retrieval options based on how quickly archived data is needed.
- Long-term retention – Support archival requirements for backups, records, and historical information.
- Automated archiving – Work with S3 Lifecycle rules to transition eligible objects between storage classes.
- Data protection – Support encryption, access controls, and retention features through Amazon S3.

References:

Amazon Web Services. (2022). *AWS Academy Cloud Foundations learner guide*. Amazon Web Services.

Amazon Web Services. (n.d.). *Amazon EBS volumes*. https://docs.aws.amazon.com/ebs/latest/userguide/ebs-volumes.html

Amazon Web Services. (n.d.). *Understanding and managing Amazon S3 storage classes*. https://docs.aws.amazon.com/AmazonS3/latest/userguide/storage-class-intro.html

Amazon Web Services. (n.d.). *Understanding archive retrieval options*. https://docs.aws.amazon.com/AmazonS3/latest/userguide/restoring-objects-retrieval-options.html

Amazon Web Services. (n.d.). *What is Amazon Elastic File System?* https://docs.aws.amazon.com/efs/latest/ug/whatisefs.html

PROPERTY OF STI
PROPERTY OF STI
PROPERTY OF STI